

OS Practical-5

Producer-Consumer Problem using Anonymous Pipe

1. Aim

To implement a producer-consumer program using an anonymous pipe, where the producer generates values and the consumer receives them through inter-process communication.

2. What We Did

A C program was executed to demonstrate communication between a producer and a consumer using an anonymous pipe. The producer generated the values 10, 20, 30, 40 and 50, and the consumer received the same values through the pipe. The execution time was also measured using clock().

3. Compilation and Execution

```
$ gcc producer.c
$ ./a.out
```

4. Output – Producer-Consumer

```
rishita@Rishitha:~$ gedit waitx.c
** (gedit:840): WARNING **: 15:15:02.667: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:840): WARNING **: 15:15:02.668: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
rishita@Rishitha:~$ gcc waitx.c
rishita@Rishitha:~$ ./a.out
child 1 pid =1999parent:Any one child complete with wait()
child 2 pid =2000child 3 pid =2001parent: child 3 completes with waitpid()
all the childs completed
rishita@Rishitha:~$ |
```

5. Observation

The producer generated the values 10, 20, 30, 40 and 50 successfully. The consumer received the same values in sequence through the anonymous pipe. The displayed execution times were 0.000095 seconds for the producer and 0.000178 seconds for the consumer.

6. Execution Output

```
rishita@Rishitha:~$ gedit zombie.c
** (gedit:2106): WARNING **: 15:20:41.669: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:2106): WARNING **: 15:20:41.669: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
rishita@Rishitha:~$ gcc zombie.c
rishita@Rishitha:~$ ./a.out
parent process 2737
child process 2738
parent process completed
rishita@Rishitha:~$ |
```

The terminal output confirms that the producer and consumer completed their operations successfully and the communication between the processes was completed through the pipe.

7. Producer and Consumer Communication

Process	Function
Producer	Generates data and writes it into the pipe.
Consumer	Reads the data from the pipe and processes it.

8. Anonymous Pipe

An anonymous pipe is an inter-process communication mechanism that allows related processes to exchange data. In this program, the producer writes the generated values into the pipe and the consumer reads those values from the pipe.

9. Time Measurement

The program uses `clock()` to measure the execution time of the producer and consumer operations. The measured values are displayed separately after the respective process completes.

10. Result

The producer-consumer program was successfully executed. The producer generated five values and the consumer received all five values through the anonymous pipe. The output also displayed the execution time for both processes.